

Calculus Pt. 1 (Derivation)

UChicago Political Science Math Camp

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Analysis of Functions

So far, we've talked about various results we might want to recover from a function:

- ▶ The value of an **output** given **inputs**
- ▶ The **inverse** (an input given an output)
- ▶ The **limit** of a function when a numerical output may not exist
- ▶ The **end behavior** of a function as a parameter approach zero, infinity, or some asymptotical border

The Plan

1. What the hell is this and why do we need this?
2. How do we actually do this?
3. Many rules, very quickly

What The Hell Is This And Why Do We Need It?

What is Calculus?

The study of how functions change

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We use two fundamental operations for calculus:

- ▶ Derivation
- ▶ Integration

What is Derivation?

- ▶ A derivative expresses the **instantaneous rate of change** of a function
- ▶ We isolate this rate to represent the impact of a single variable¹
- ▶ *How does the function's output change as a given input variable changes?*

Take the derivative of f with respect to $x \rightarrow$

How does a change in x affect the value of f ?

¹mostly (don't worry about it) (for now)

Derivation in Political Science

We use derivatives whenever we want to evaluate *marginal effects* – that is, almost always

For example, we might ask: what is the effect of educational attainment on future wealth? Of legislature size on legislative productivity? Of polarization on political violence?

Any causal mechanism essentially requires this approach – verbally, mathematically, or otherwise

An Example: Legislative Capacity

Suppose we have some function $f : [0, 1] \rightarrow \mathbb{R}_+$ which transforms an index of political polarization to an expected number of bills passed per session.

Derivation lends two possible conclusions:

1. The *direction* of the effect of polarization on legislative capacity, and
2. The *magnitude* of this effect

Why might direction matter more than magnitude?

What Do Derivatives Look Like?

A derivative can look like one of two outputs:

1. A **function** $f'(x)$, which returns the value of a derivative for some input $x \in \mathbb{R}$
2. A **value** $f'(x = a)$, the output value of f' for a specific $x = a \in \mathbb{R}$

Usually, we want the function $f'(x)$ unless specifically given some $x = a$

Differentiable Functions

A function is **differentiable** if it can be derived

A differentiable function is **smooth** in appearance *and has no vertical asymptotes*.

$$\implies f(x) \in \mathbb{R} \text{ and } f(x) \neq \pm\infty$$

Derivatives & Notation

(From above) Take the derivative of f with respect to $x \rightarrow$ How does a change in x affect the value of f ?

- ▶ $\frac{df}{dx}$, often shortened to $\frac{d}{dx}$
- ▶ $\frac{\partial f}{\partial x}$, sometimes shortened to $\frac{\partial}{\partial x}$
- ▶ $f'(x)$ when there is a single or very clear choice variable
- ▶ f_x in some economics literature (usually there is a comment)

For large equations, you might use this notation as an operator:

$$\frac{\partial}{\partial x}(\text{write function here})$$

Legislative Capacity Example

Suppose f takes the following form:

$$f(x) = \frac{1}{x}$$

And (just trust me for now) f' is as follows:

$$f'(x) = -\frac{1}{x^2}$$

Where $x \in [0, 1]$ represents our polarization index.

- ▶ What direction is the (supposed) relationship between polarization and capacity?
- ▶ When is the marginal effect of polarization on capacity stronger / weaker?

Units of Derivation

As we alluded earlier, derivation invites a unique consideration: what units does a derivative represent?

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A derivative $\frac{d}{dx}f(x)$ the unit rate of change of f with respect to a single-unit increase in x , but makes no further clarification as to either unit.

How might we overcome this to compare relative magnitudes?

Can You Take A Derivative's Derivative?

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yes

Iterated Derivatives

Any differentiable function can be derived (including the derivation of a function):

$$\frac{d}{dx} f(x) = \frac{df(x)}{dx} = f'(x)$$

$$\frac{d}{dx} \frac{df(x)}{dx} = \frac{d^2 f(x)}{dx} = f''(x)$$

$$\frac{d}{dx} \frac{d^2 f(x)}{dx} = \frac{d^3 f(x)}{dx} = f'''(x) \dots$$

$$\frac{d}{dx} \frac{d^n f(x)}{dx} = \frac{d^{n+1} f(x)}{dx}$$

Iterated Derivatives

We refer to the 2nd, 3rd, ... n th derivative in this sense.

- ▶ The derivative of $f(x)$ represents the rate of change of $f(x)$.
- ▶ The second derivative of $f(x)$ is $f'(x)$, and represents the rate of change of the rate of change of $f(x)$...
- ▶ The easiest example is physics, where $t = \text{time}$:

$$\text{Position} \Rightarrow \frac{d}{dt} = \text{Velocity} \Rightarrow \frac{d}{dt} = \text{Acceleration} \Rightarrow \frac{d}{dt} = \text{Jerk}$$

Integration simply reverses the flow!

Beating A Dead Horse

We go back to our legislative capacity example, and continue to extent:

$$f(x) = \frac{1}{x}$$

$$f'(x) = -\frac{1}{x^2}$$

$$f''(x) = \frac{2}{x^3}$$

The rate of change increases over time – in direction, but NOT in magnitude.

Broader Questions

Does the n th derivative of a (differentiable) function always exist?

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Does the n th derivative of a (differentiable) function always exist?

Does the n th derivative return interpretable units of measurement?

Two Trivial Cases

There are two types of functions² with trivial derivatives:

1. Constant functions: $f(x) = 2$
2. Linear functions: $f(x) = 2x$

We can revert to our trusty slope calculations:

$$y = mx + b \implies \frac{d}{dx} = m$$

$$m = \frac{\text{Rise}}{\text{Run}} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

²(1) is a special case of (2)

Getting Serious

Of course, we are rarely interested in trivial derivatives (OLS coefficients aside)...

To better approach the *instantaneous* rate of change for a non-linear function, we want to shrink our “Run,” or Δx

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What might this look like?

The smaller the “Run,” the closer the approximation... What might this imply?

The Limit Definition

$$\frac{df}{dx} = \lim_{h \rightarrow 0} \frac{\overbrace{f(a+h)}^{y_2} - \overbrace{f(a)}^{y_1}}{\underbrace{h}_{x_2 - x_1}}$$

...is mostly bullshit

Analytically, this makes sense, but a huge problem pops up right away...

Sometimes we can solve through this anyways. Take $f(x)$, and $x = 2$:

$$f'(2) = \lim_{h \rightarrow 0} \frac{(2+h)^2 - 2^2}{h}$$

$$\frac{(2+h)^2 - h^2}{h} \Rightarrow \frac{4 + 4h + h^2 - 4}{h} \Rightarrow \frac{4h + h^2}{h} \Rightarrow \frac{h(4+h)}{h} \Rightarrow 4 + h$$

$$\lim_{h \rightarrow 0} 4 + h = 4$$

$$\therefore f'(2) = 4$$

What about $f(x) = (1-x)^2$? Or a multi-term polynomial? Or a generalizable function?

How Do We Actually Do This?



(maybe)

The Good Stuff

We can use a series of rules instead of the limit definition to derive a function in its general form for any value of x :

$$f(x) \implies [\text{RULE}] \implies f'(x)$$

Note that now, our derivative is itself another function of x , and requires no specification of x or a :

$$f'(x) = \frac{d}{dx} f(x) \quad \forall x \in \text{domain of } f$$

(Wanna play with notation here?)

98% of Mathematical History

The **Power Rule**:

$$\frac{d}{dx}x^a = a \cdot x^{a-1}$$

Power Rule Examples

- ▶ A simple application

$$\frac{d}{dx}x^2 = 2 \cdot x^{2-1} = 2x$$

- ▶ Works with any natural numbers ($\mathbb{N}$)

$$\frac{d}{dx}x^5 = 5 \cdot x^{5-1} = 5x^4$$

- ▶ Also works with negative numbers ($\mathbb{Z}$)

$$\frac{d}{dx} \frac{1}{x^3} = \frac{d}{dx} x^{-3} = -3 \cdot x^{-4} = -\frac{3}{x^4}$$

- ▶ Also works with roots ($\mathbb{Q}$ and by expansion $\mathbb{R}$)

$$\frac{d}{dx} \sqrt[3]{x^5} = \frac{d}{dx} x^{\frac{5}{3}} = \frac{5}{3} \cdot x^{\frac{5}{3}-1} = \frac{5}{3} x^{\frac{2}{3}} = \frac{5\sqrt[3]{x^2}}{3}$$

This will surely create no problems for us in the future

The Sum Rule

$$\frac{d}{dx} \left(f(x) + g(x) \right) = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$$

that's it

A Power Rule Corollary

1. We know that $\frac{d}{dx}x^a = a \cdot x^{a-1}$ (**Power Rule**) and that $\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}f(x) + \frac{d}{dx}g(x)$ (**Sum Rule**).
2. We also know that for a constant b , $b \cdot c = \sum_1^b c$, or c added b times.
3. Then, the derivative is the sum of x^a added up b times:
$$\frac{d}{dx} \left[b \cdot x^a \right] = \sum_1^b \frac{d}{dx} x^a$$
4. We know that $\sum_1^n x = nx$, so

$$\sum_1^b \frac{d}{dx} x^a = b \cdot \frac{d}{dx} x^a \quad \implies \quad \frac{d}{dx} \left[b \cdot x^a \right] = b \cdot a \cdot x^{a-1}$$

$\implies$ **Coefficients:** The coefficient of a term is kept as a coefficient of that term's derivative. (i.e., multiply a by the coefficient as well)

And Another One

1. We know that $x^0 = 1 \forall x$.
2. Therefore, $a = a \cdot x^0$ for any $a \in \mathbb{R}$.
3. So, the derivative of a can be solved using the power rule:

$$\frac{d}{dx}a = \frac{d}{dx}\left[a \cdot x^0\right] = 0 \cdot a \cdot x^{-1} = 0$$

$\implies$ **Constants:** The derivative of a constant is 0.

Working In Terms

Always work with the Power Rule when possible!

We can use the sum rule to enable the power rule, so long as we can expand expressions (do this if possible!):

$$2x(x - 1) \implies 2x^2 - 2x$$

$$(x - 1)(x + 4) \implies x^2 + 3x - 4$$

But, what if we can't?

Many Rules, Very Quickly (The Important Ones)

The Other 2%

The **Chain Rule**:

$$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$$

or

$$\frac{d}{dx}f(u) = f'(u) \cdot u'$$

Depending on your persuasion

Chain Rule Examples



$$\frac{d}{dx}(x-1)^2$$

Set $u = x - 1$ and $f(u) = u^2$

Chain Rule Examples



$$\frac{d}{dx}(x-1)^2$$

Set $u = x - 1$ and $f(u) = u^2$

▶ Then, $u' = 1$ and $f'(u) = 2u$

Chain Rule Examples



$$\frac{d}{dx}(x-1)^2$$

Set $u = x - 1$ and $f(u) = u^2$

▶ Then, $u' = 1$ and $f'(u) = 2u$

▶ Put it together, then

$$f'(u) \cdot u' = 2u \cdot 1 = 2(x-1)$$

$$\frac{d}{dx}(x-1)^2 = 2(x-1)$$

Note that this would be easier to work with as $(x-1)^2 = x^2 - 2x + 1$, though...

More Chain Rule

$$\frac{d}{dx} \frac{1}{(2x^2 + 3x)^4}$$

We could expand...

$$\frac{1}{(2x^2 + 3x)^4} = \frac{1}{(2x^2 + 3x)(2x^2 + 3x)(2x^2 + 3x)(2x^2 + 3x)}$$

And then derive a 12 term expression. Or:

More Chain Rule

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$$\frac{1}{(2x^2 + 3x)^4} = \frac{1}{(2x^2 + 3x)(2x^2 + 3x)(2x^2 + 3x)(2x^2 + 3x)}$$

And then derive a 12 term expression. Or:

$$u = 2x^2 + 3x, \quad f = \frac{1}{u^4} = u^{-4}$$

$$u' = 4x + 3, \quad f'(u) = -4u^{-5} = -\frac{4}{u^5}$$

$$\frac{d}{dx} f(u) = f'(u) \cdot u' = \frac{-4(4x + 3)}{(2x^2 + 3x)^5}$$

A Chain Rule Warning

- ▶ With polynomials (incl. roots), the chain rule is a **helpful tool**
- ▶ With non-polynomials (i.e. exponential or trigonometric) or unspecified/implicit functions, the chain rule can become a **necessary step**

Product Rule

- ▶ Sometimes, we need to derive multiple functions housed within one term (i.e., operations are multiplied rather than added)
- ▶ If we can't expand algebraically (or if doing so is prohibitively complicated), we can use the **product rule**:

$$\frac{d}{dx} \left[u(x) \cdot v(x) \right] = u'(x) \cdot v(x) + u(x) \cdot v'(x)$$

Product Rule Example

$$(2x^2 + 3)(x^4 + 4x)$$

Product Rule Example

Set $u(x) = 2x^2 + 3$ and $v(x) = x^4 + 4x$. Then,

$$(2x^2 + 3) \cdot (x^4 + 4x) = u(x) \cdot v(x)$$

$$\begin{aligned}\frac{d}{dx} [u(x) \cdot v(x)] &= u'(x) \cdot v(x) + u(x) \cdot v'(x) \\ &= \frac{d}{dx} [2x^2 + 3] \cdot (x^4 + 4x) + (2x^2 + 3) \cdot \frac{d}{dx} [x^4 + 4x] \\ &= 4x(x^4 + 4x) + (2x^2 + 3)(4x^3 + 4)\end{aligned}$$

A Note on Simplification

Simplify with *purpose*.

Things to (often) look for:

- ▶ Reduction of terms
- ▶ Simplification of fractions
- ▶ Combining exponents ($x^a x^b = x^{a+b}$, $x^{a^b} = x^{ab}$ etc.)
- ▶ Single $\pm$ constants
- ▶ Explicitly $\pm$ components (3 terms or fewer!)

More Product Rule Examples

Like the Chain Rule, the Product Rule ranges from “inefficient” to “useful” to “necessary”:

$$\frac{d}{dx} \left[(2x^2 + 3)(x^4 + 4x) \right] \implies \text{INEFFICIENT}$$

$$\frac{d}{dx} \left[(4x^2 + 2)^3(3x - x^3) \right] \implies \text{USEFUL}$$

$$\frac{d}{dx} \left[2^x(2x + x^4) \right] \implies \text{NECESSARY}$$

Many Rules, Very Quickly (The
Other Ones)

Quotient Rule

We can derive the Quotient Rule from the Product Rule, using the definition of multiplication:

$$u(x) \div v(x) = \frac{u(x)}{v(x)} = u(x) \cdot (v(x)^{-1})$$

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Then, we have:

$$\begin{aligned}\frac{d}{dx} \left[\frac{u(x)}{v(x)} \right] &= u(x) \cdot \frac{d}{dx} v(x)^{-1} + \frac{d}{dx} u(x) \cdot v(x)^{-1} \\&= u(x) \cdot \frac{-v'(x)}{(v(x))^2} + u'(x) \cdot \frac{1}{v(x)} \\&= \frac{-v'(x) \cdot u(x)}{(v(x))^2} + \frac{u'(x)}{v(x)} \\&= \frac{-v'(x) \cdot u(x)}{(v(x))^2} + \frac{u'(x) \cdot v(x)}{(v(x))^2} = \frac{u'(x) \cdot v(x) - u(x) \cdot v'(x)}{(v(x))^2}\end{aligned}$$

Quotient Rule (Sans Fluff)

$$\frac{d}{dx} \left[\frac{u(x)}{v(x)} \right] = \frac{u'(x) \cdot v(x) - u(x) \cdot v'(x)}{(v(x))^2}$$

Even if you do not remember/replicate this formula, it is equivalent to the product rule

A Notation Note

Sometimes, it's not quite clear what is being derived:

$$\frac{d}{dx} f(x) + g(x)$$

$$\frac{d}{dx} h(x)j(x)$$

(or it just looks ugly)

When in doubt, use brackets:

$$\frac{d}{dx} \left[h(x) \right] \cdot j(x) \neq \frac{d}{dx} \left[h(x)j(x) \right]$$

Practice Problems

do a few

Deriving Exponentials

Any time the deriving variable is in an exponent, the rule is a bit specific:

$$\frac{d}{dx} a^x = a^x \ln(a)$$

How might we apply the chain rule in the exponent?

Something of Note

The exponential rule is true for all positive reals. But, recall our definition that $\ln a = \log_e a$ and the logarithm rules:

$$\log_b x = y \quad b^y = x$$

We can then see the following:

$$\frac{d}{dx} e^x = e^x$$

Lastly, Logarithms

Lastly, we have the general derivation for logarithms:

$$\frac{d}{dx} \log_b x = \frac{1}{x \ln b}$$

This solves a key limitation of the power rule...

Why Logarithms Matter Here

The power rule lends that $\frac{d}{dx} nx^a = anx^{a-1}$.

If integration is the reverse of derivation (more on this Wednesday), then suppose $\int nx^a dx = \frac{n}{a+1} x^{a+1}$.

How might we then express $\int \frac{1}{x} dx$?